

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Semrau

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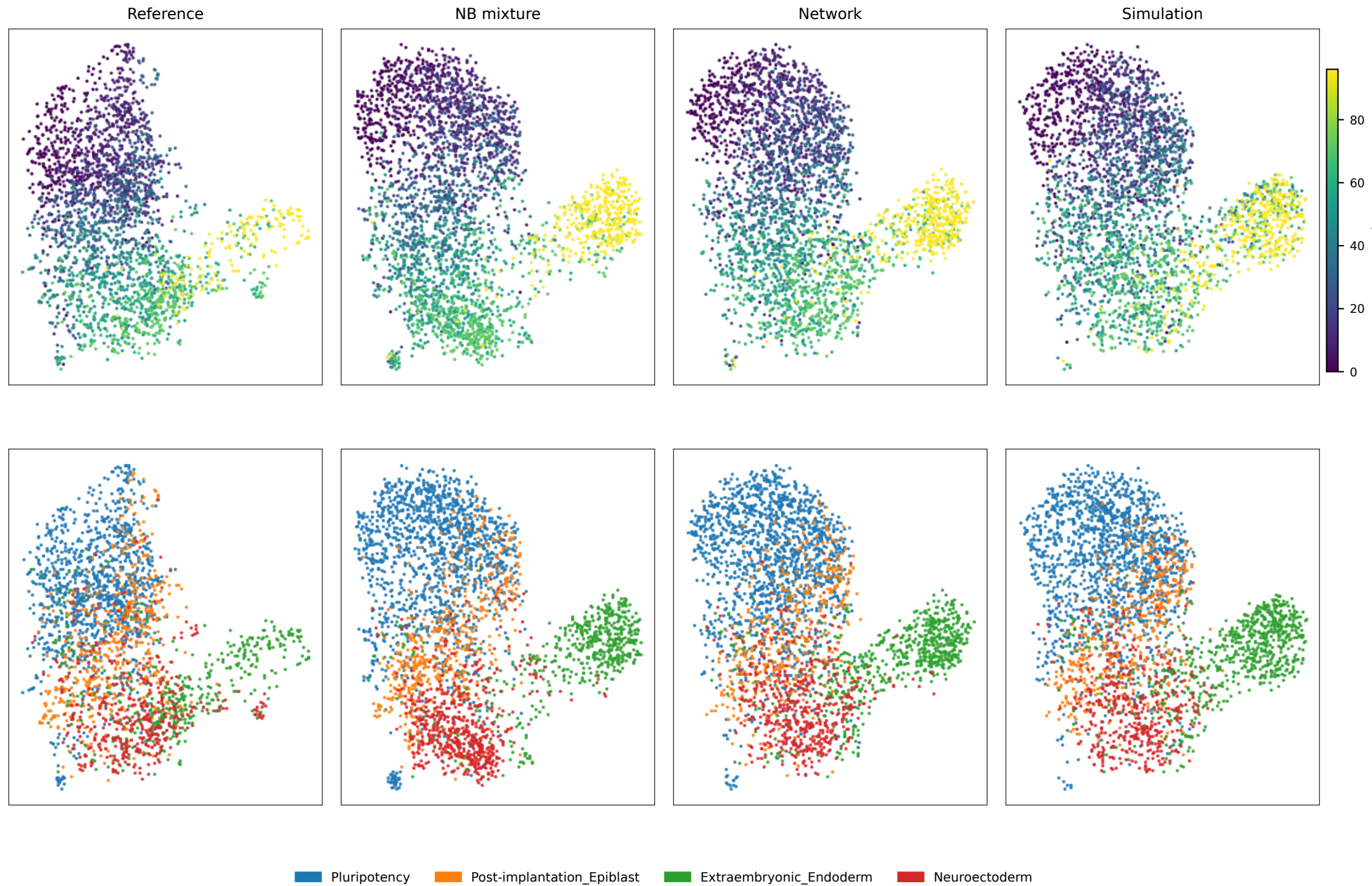
Run		Perturbations (Data/KO_OV_simulate.txt)
Split	full	✓ OV Dnmt3a
Stimulus penalisation	1.0	✓ OV Col4a2
Prior penalisation	1.0	✓ OV Hoxb2
Genes	41	
Stimuli	1	
Observed cells	2449	
Timepoints	0, 6, 12, 24, 36, 48, 60, 72, 96	
Cell types	Pluripotency, Post-implantation_Epiblast, Extraembryonic_Endoderm,	
Network shown	Neuroectoderm, network #0 / 1	
UMAP preprocessing	normalise=False, log1p=True	

Contents

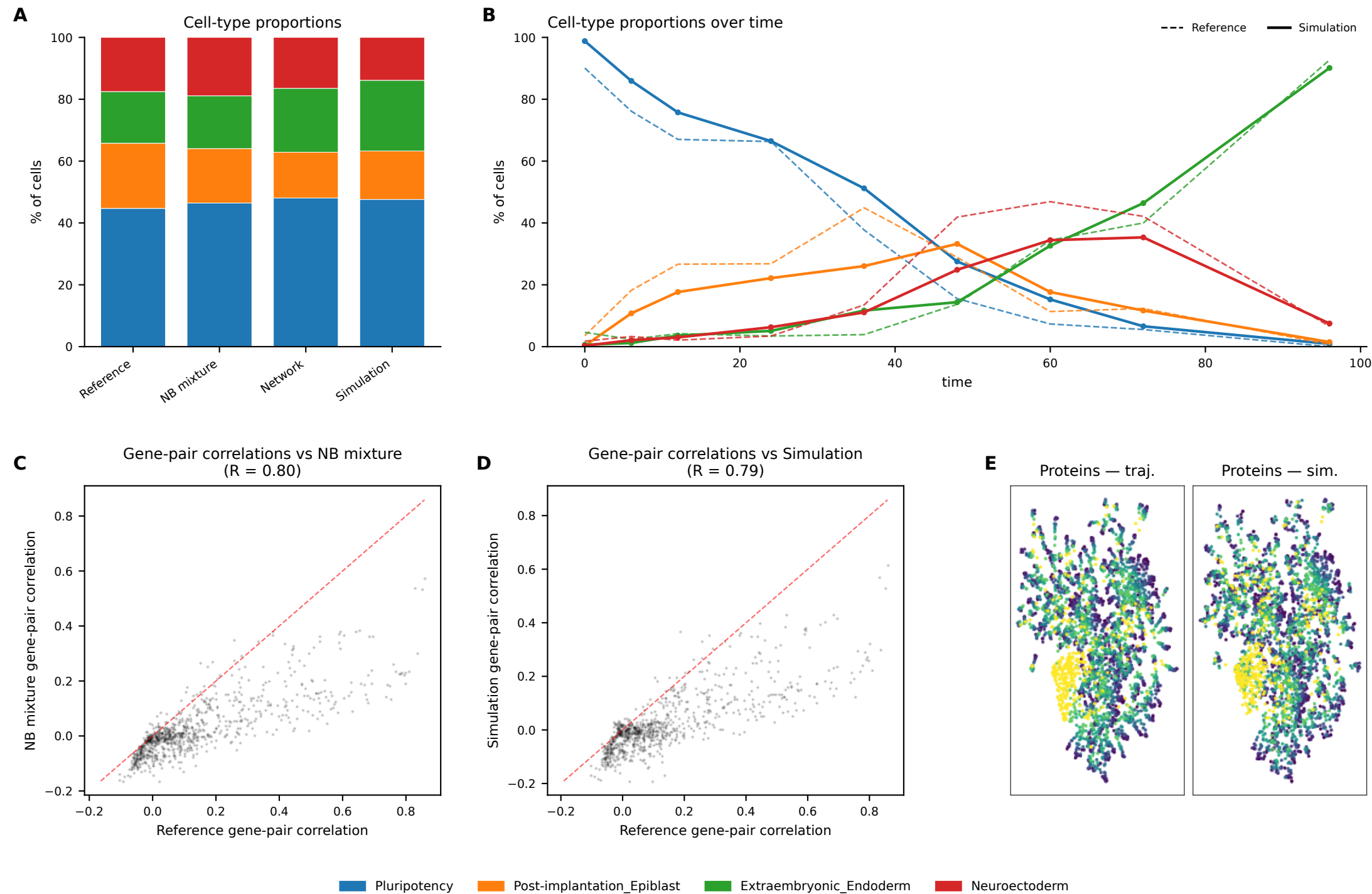
1. Generative model — UMAPs of data, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots) and top-10 regulators + stimulus
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth
5. Learned dynamics — mRNA and protein velocity fields (mechanistic and along trajectories)

1. Generative model — trajectories and simulation

Joint UMAP of the observed data (Reference), NB mixture, network-driven modes and full simulation.



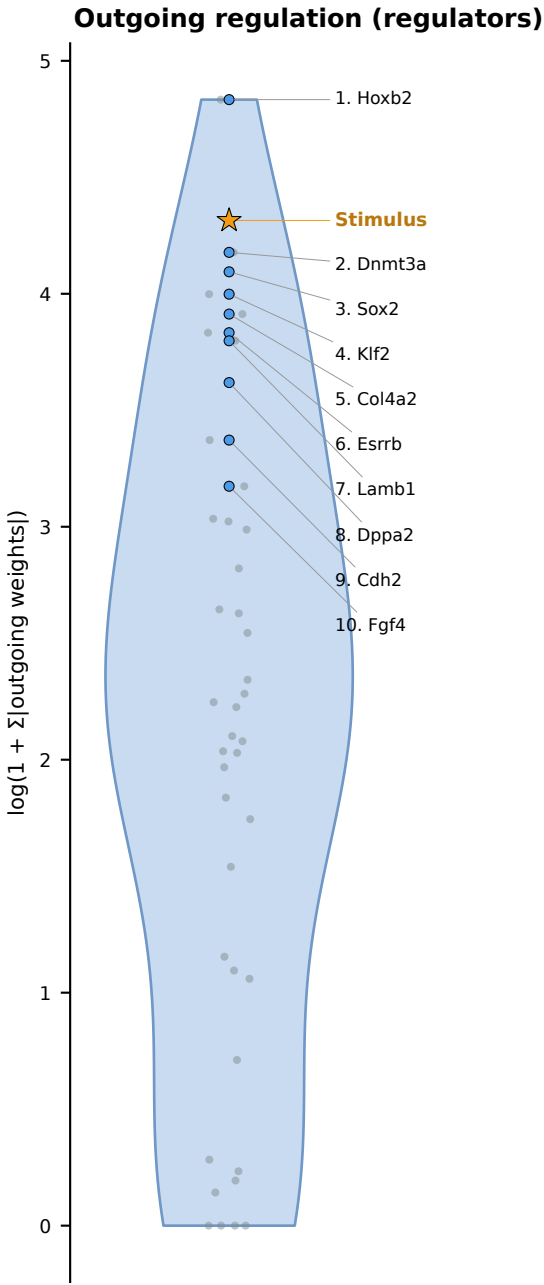
1. Generative model — quantitative checks



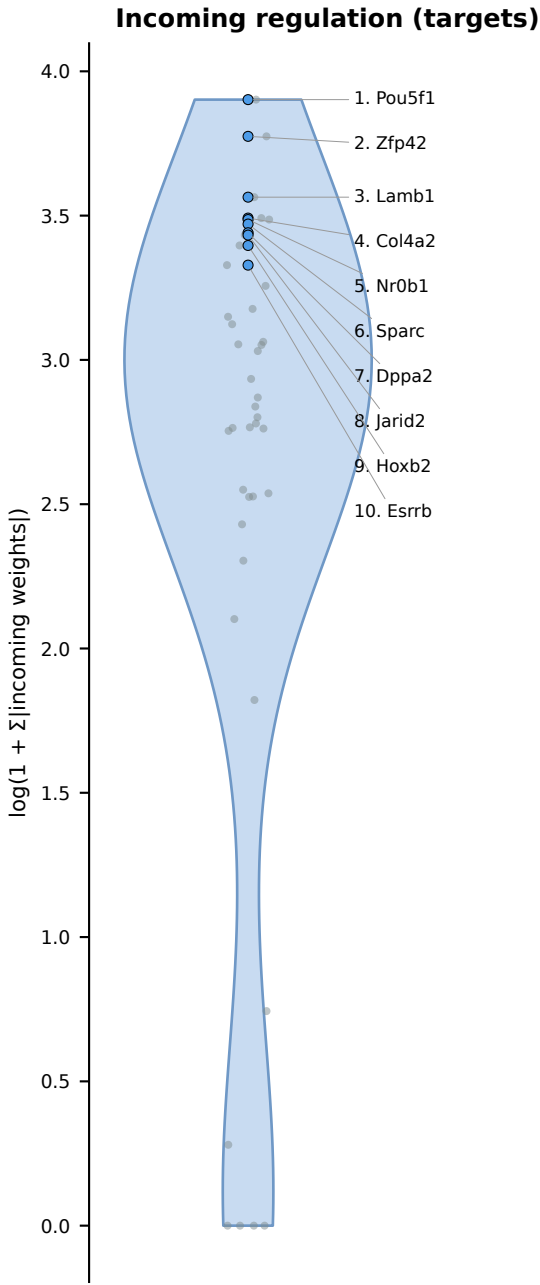
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 10 genes labelled; stimulus shown as a star (stimulus = 1.0 $\geq$ 0.5).

A



B



C

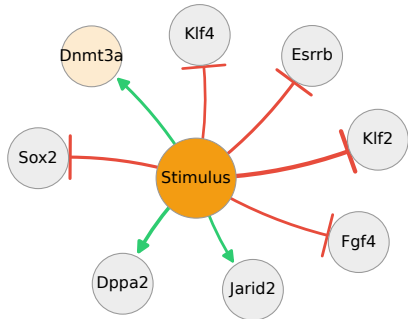
Top regulators

#	Regulator	$\Sigma w $	# act.	# inh.
1	Hoxb2	124.64	20	11
2	Dnmt3a	64.22	10	24
3	Sox2	58.99	17	9
4	Klf2	53.51	24	10
5	Col4a2	49.04	12	8
6	Esrrb	45.20	16	14
7	Lamb1	43.62	14	6
8	Dppa2	36.30	14	20
9	Cdh2	28.14	10	7
10	Fgf4	22.90	23	11
★	Stimulus	73.80	20	9

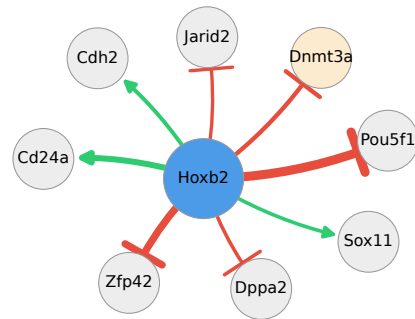
2. Gene regulatory network — top regulators and their main targets

Top 8 targets per regulator; edge width $\propto$ |weight| (global scale). Perturbed genes (KO/OV) are highlighted in beige.

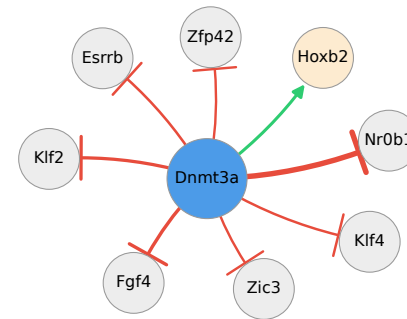
Stimulus



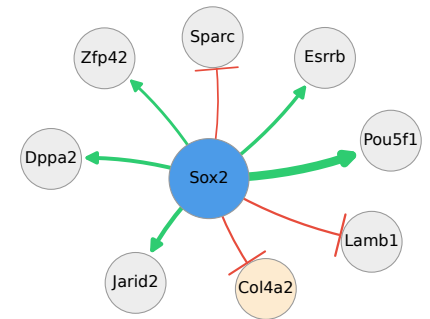
#1 Hoxb2



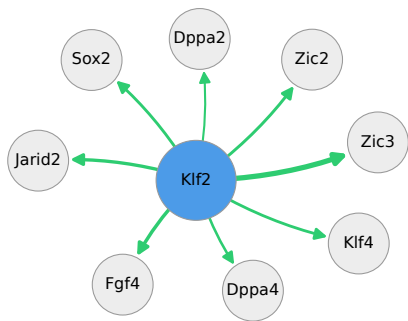
#2 Dnmt3a



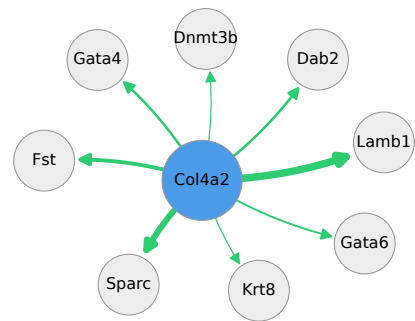
#3 Sox2



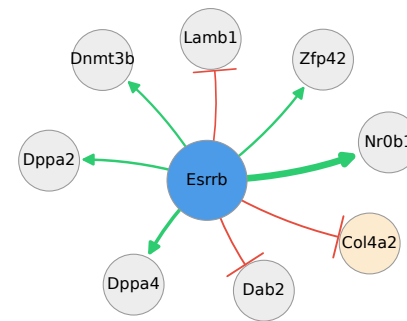
#4 Klf2



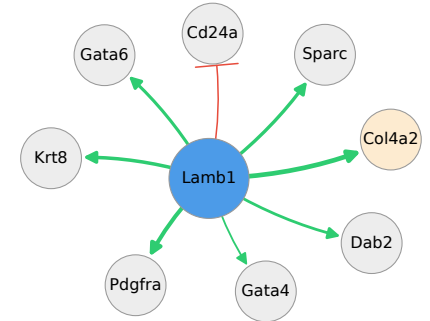
#5 Col4a2



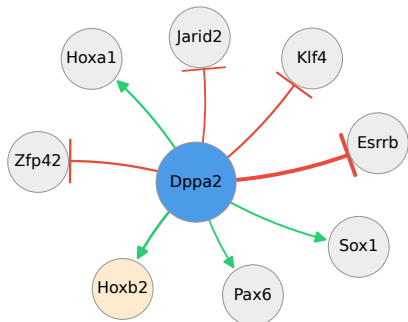
#6 Esrrb



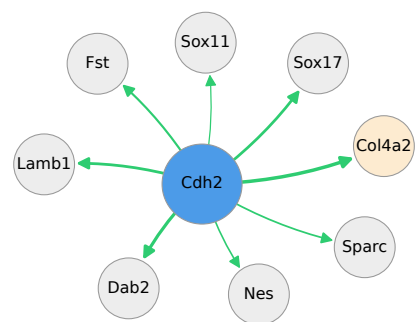
#7 Lamb1



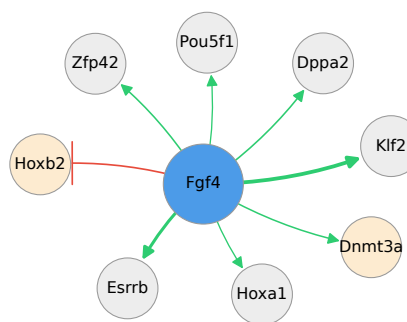
#8 Dppa2



#9 Cdh2



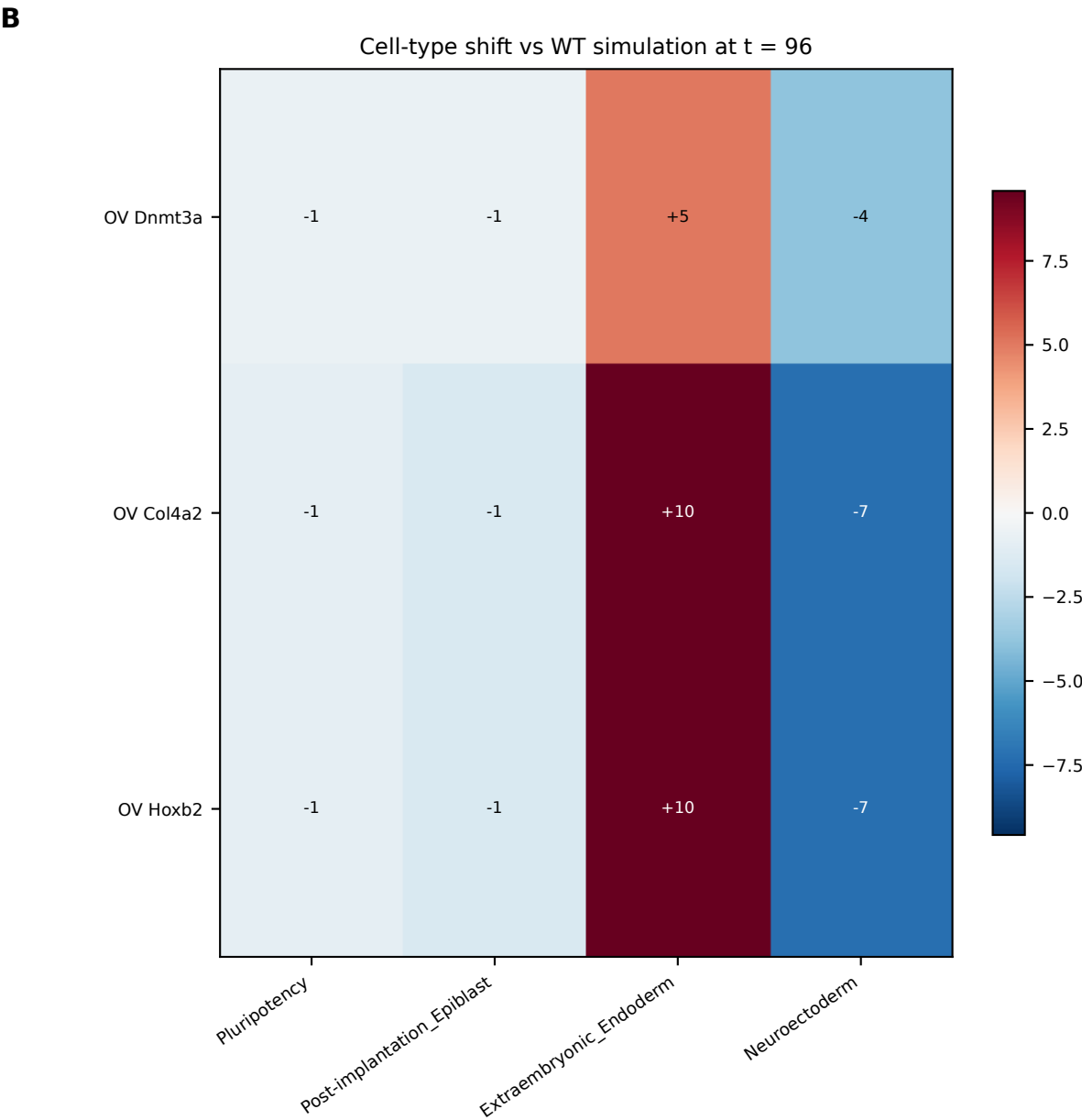
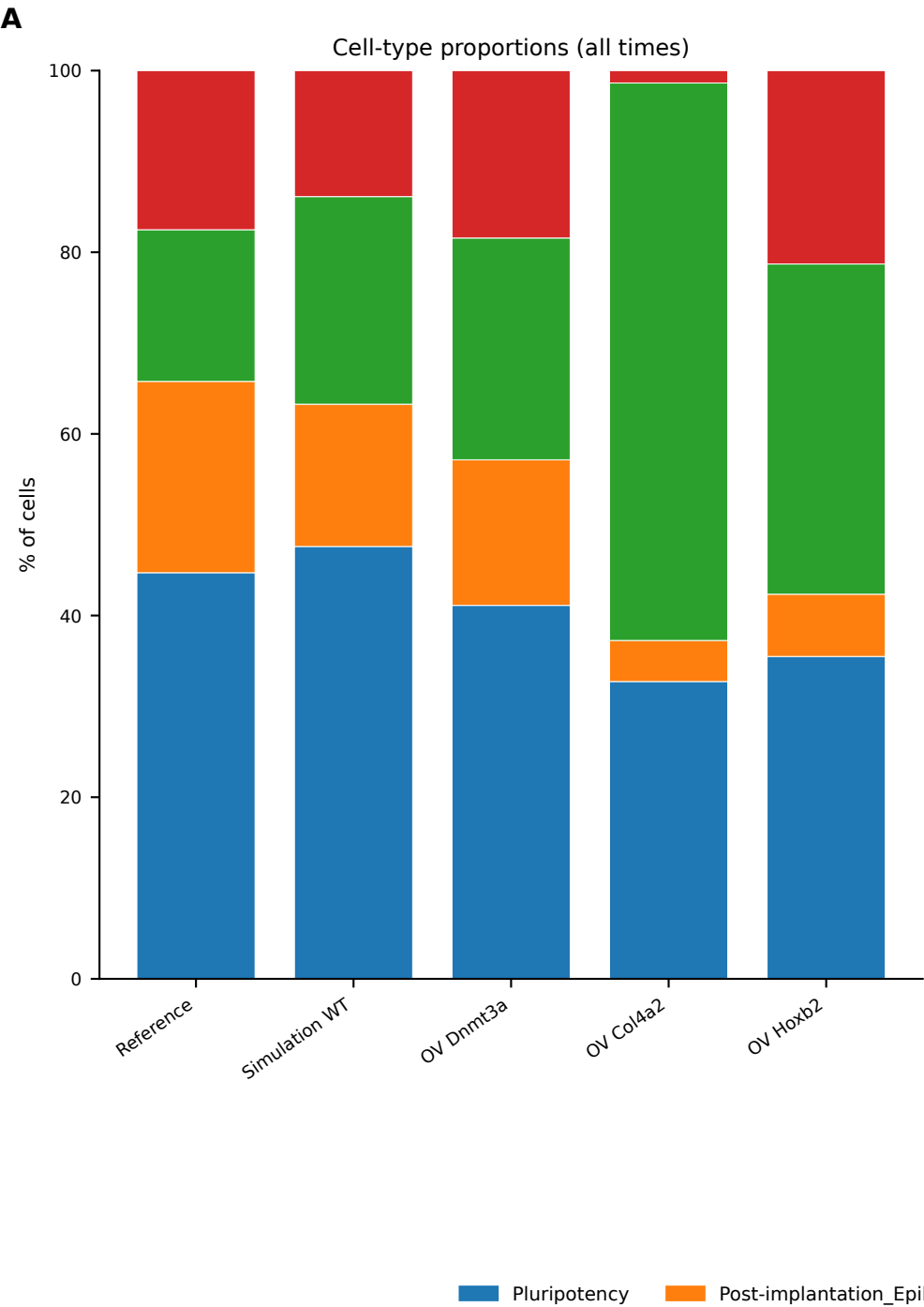
#10 Fgf4



→ Activation
 → Inhibition
 ● Top regulator
 ● Stimulus

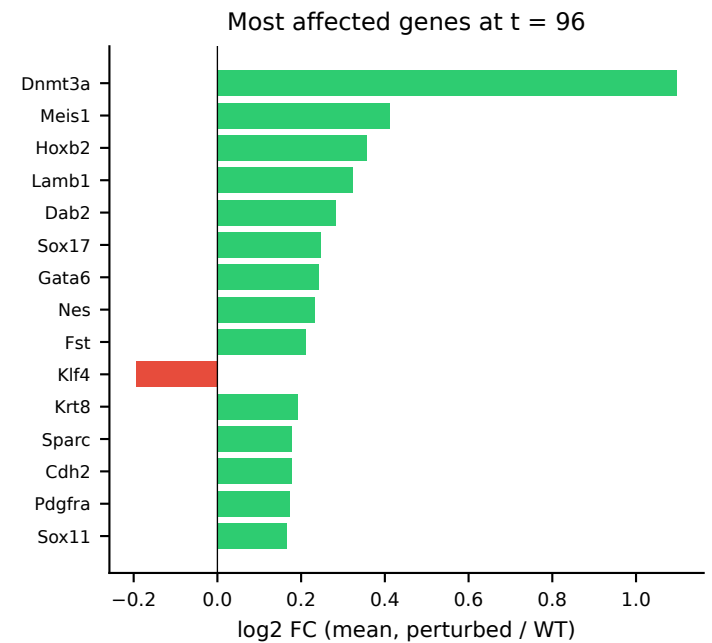
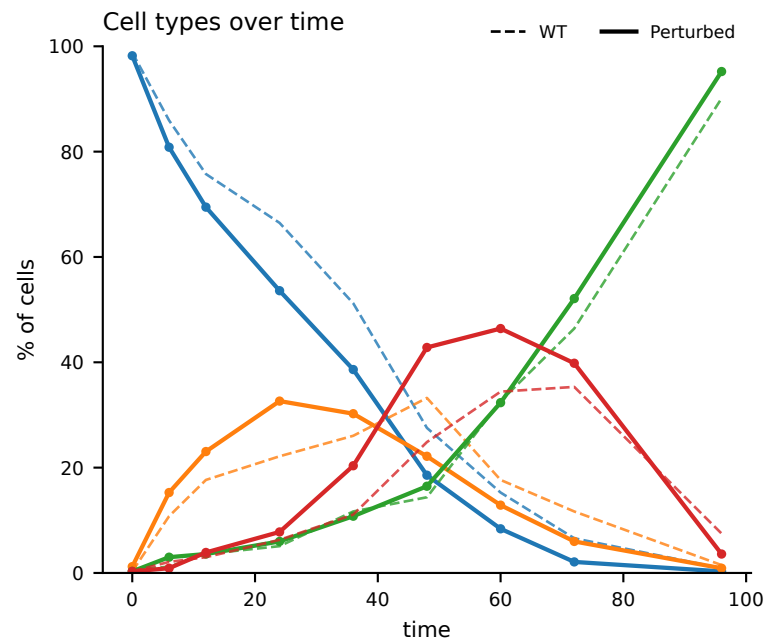
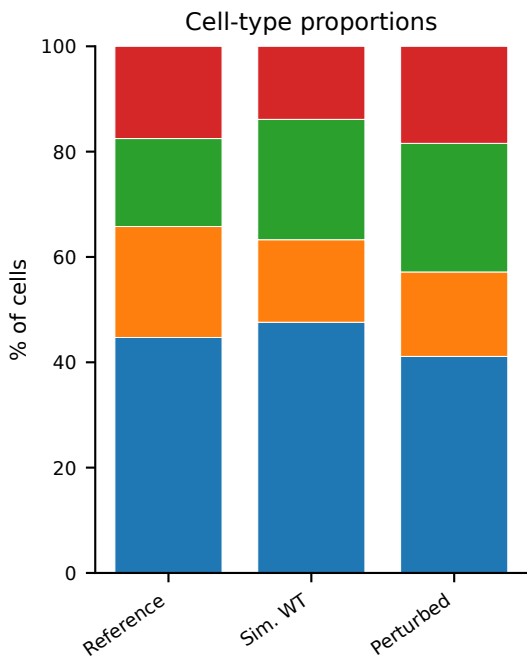
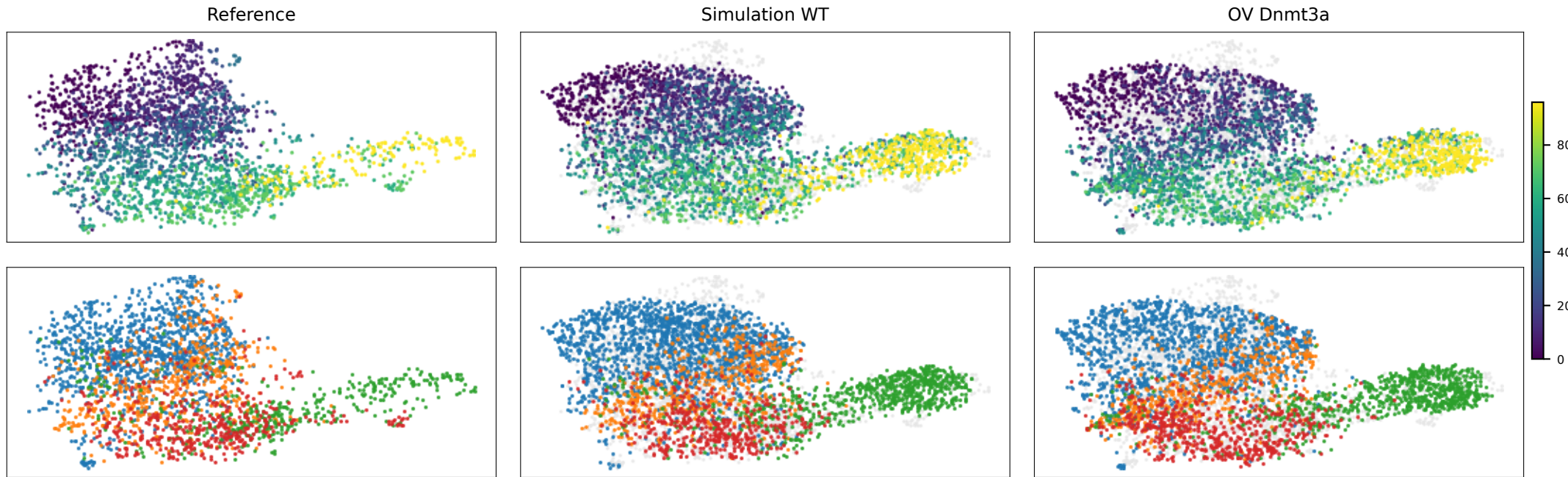
3. In-silico perturbations — overview

Cell types predicted by a random forest trained on the observed data.



3. Perturbation — OV Dnmt3a

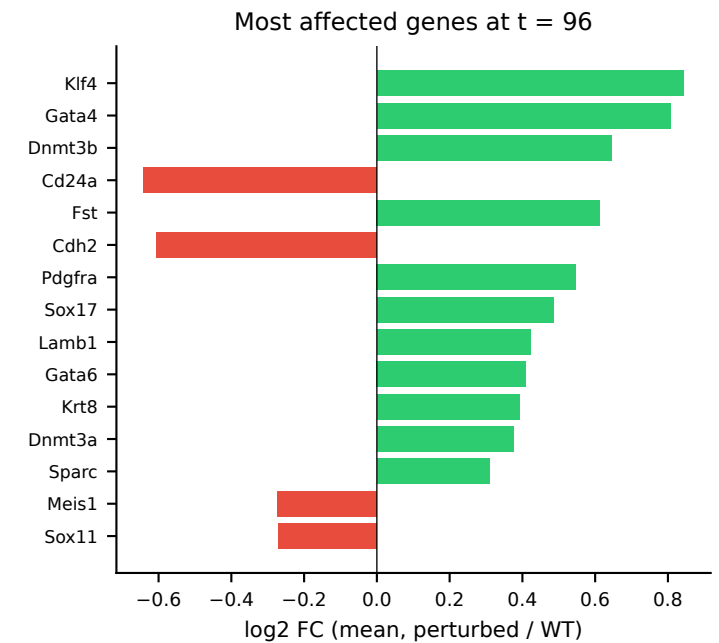
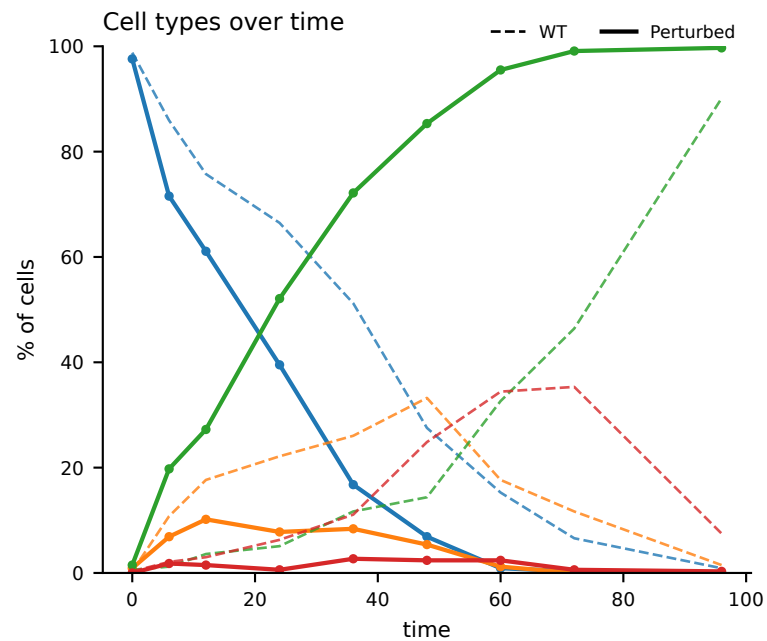
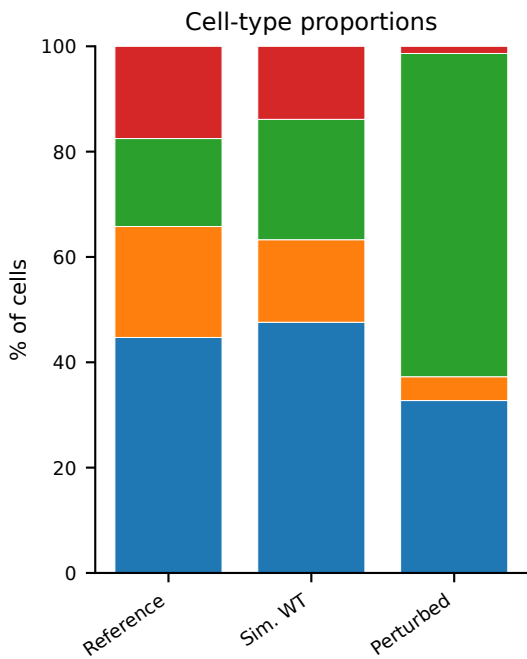
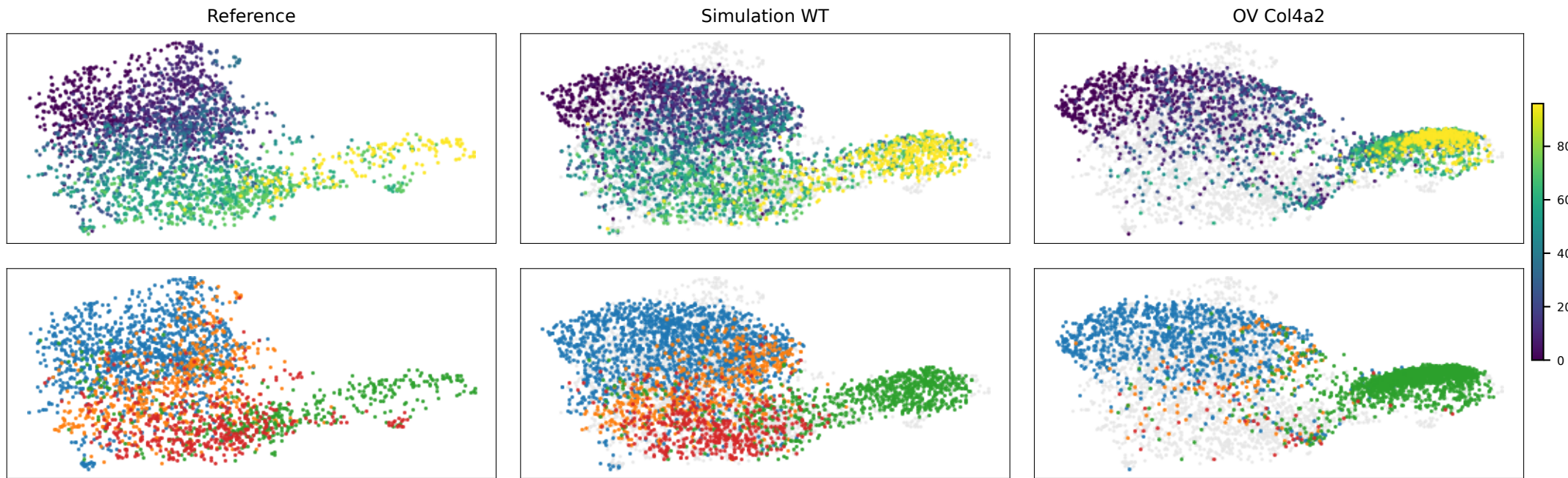
KO_none_OV_Dnmt3a. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



Pluripotency Post-implantation_Epiblast Extraembryonic_Endoderm Neuroectoderm

3. Perturbation — OV Col4a2

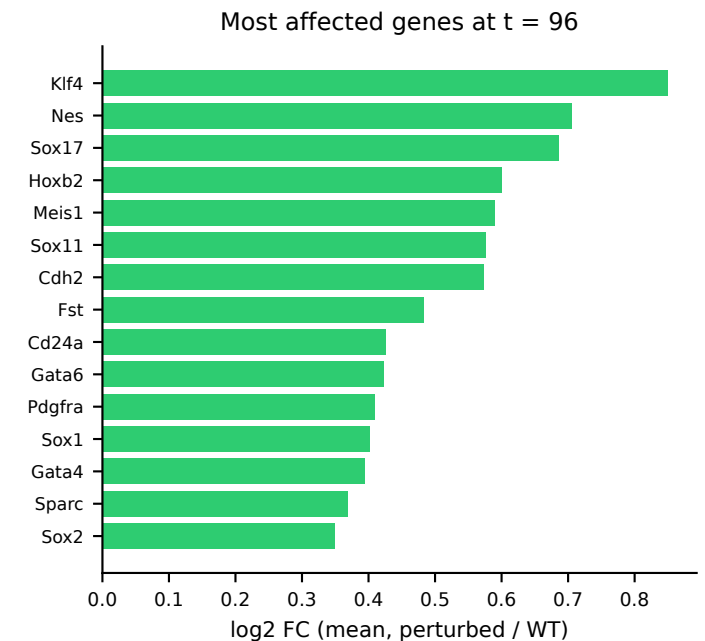
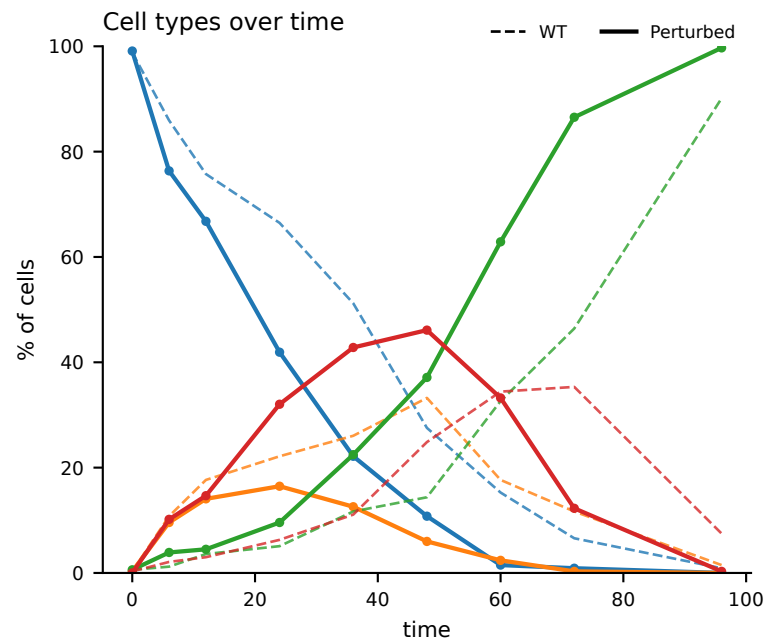
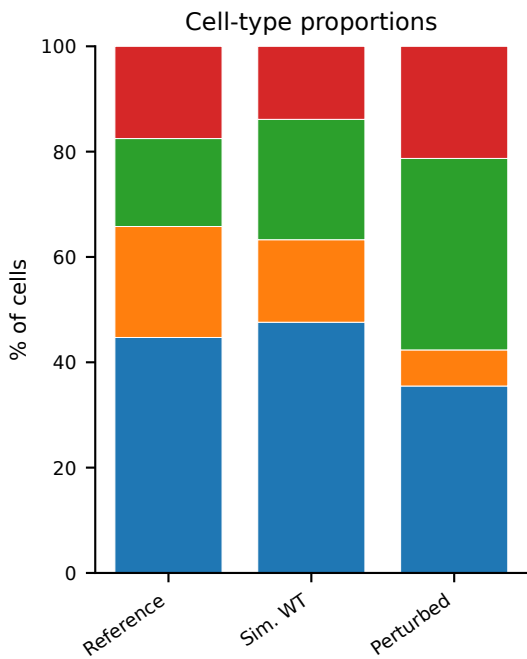
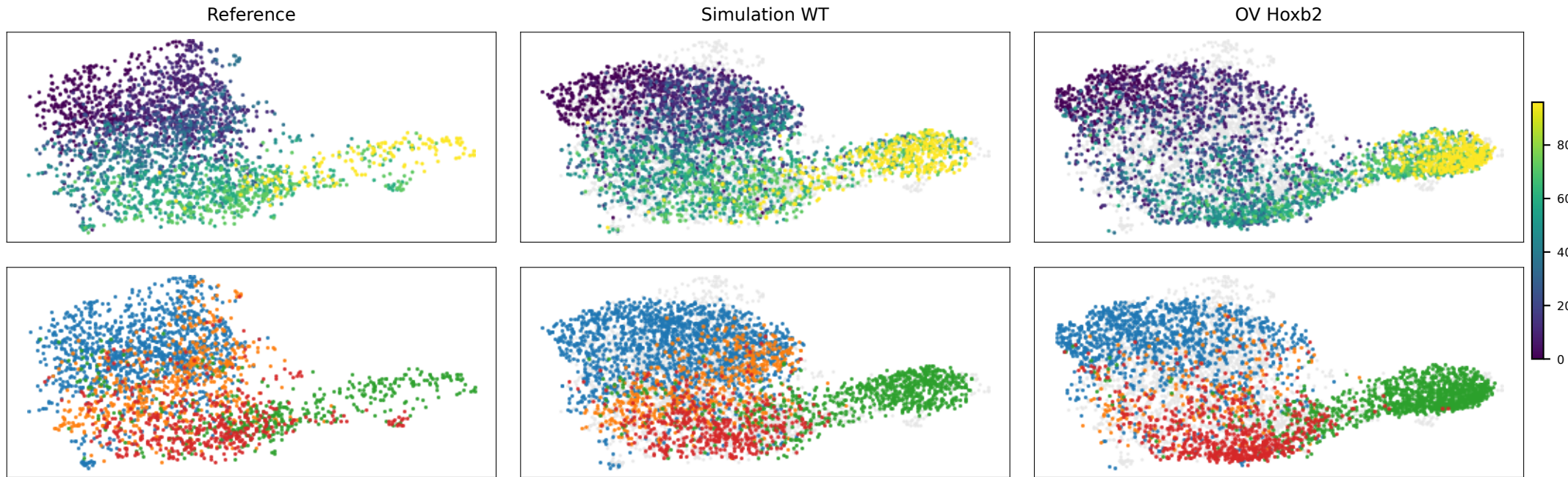
KO_none_OV_Col4a2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



Pluripotency Post-implantation_Epiblast Extraembryonic_Endoderm Neuroectoderm

3. Perturbation — OV Hoxb2

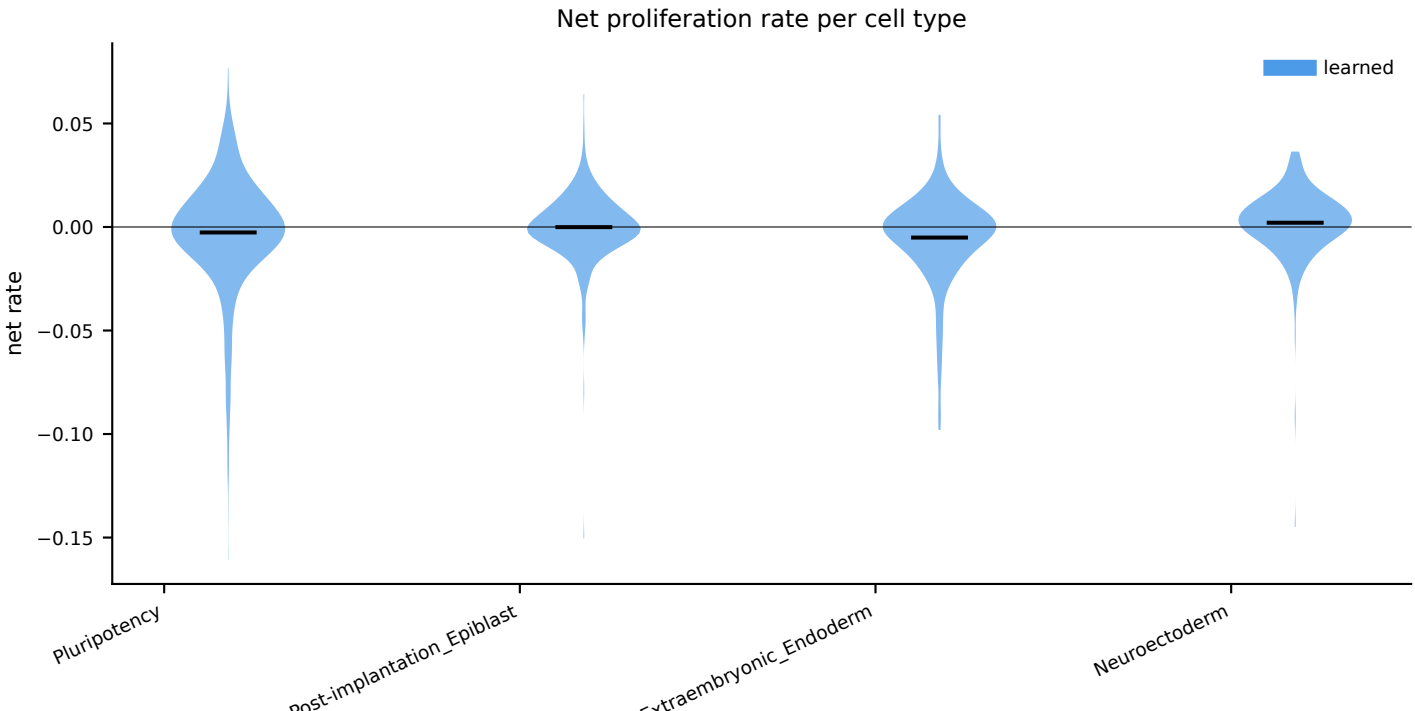
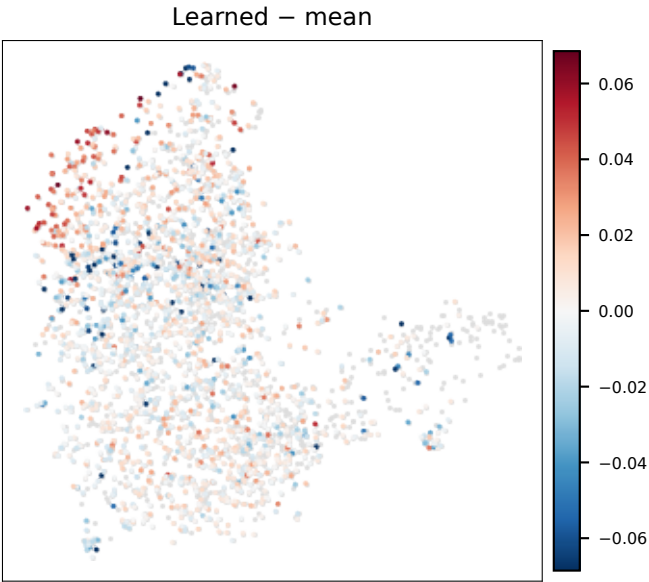
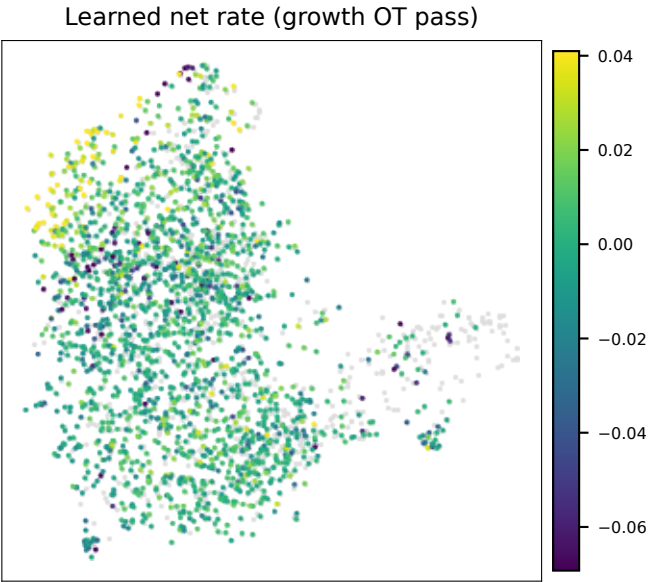
KO_none_OV_Hoxb2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



4. Proliferation — prior vs learned net rate (birth – death)

Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

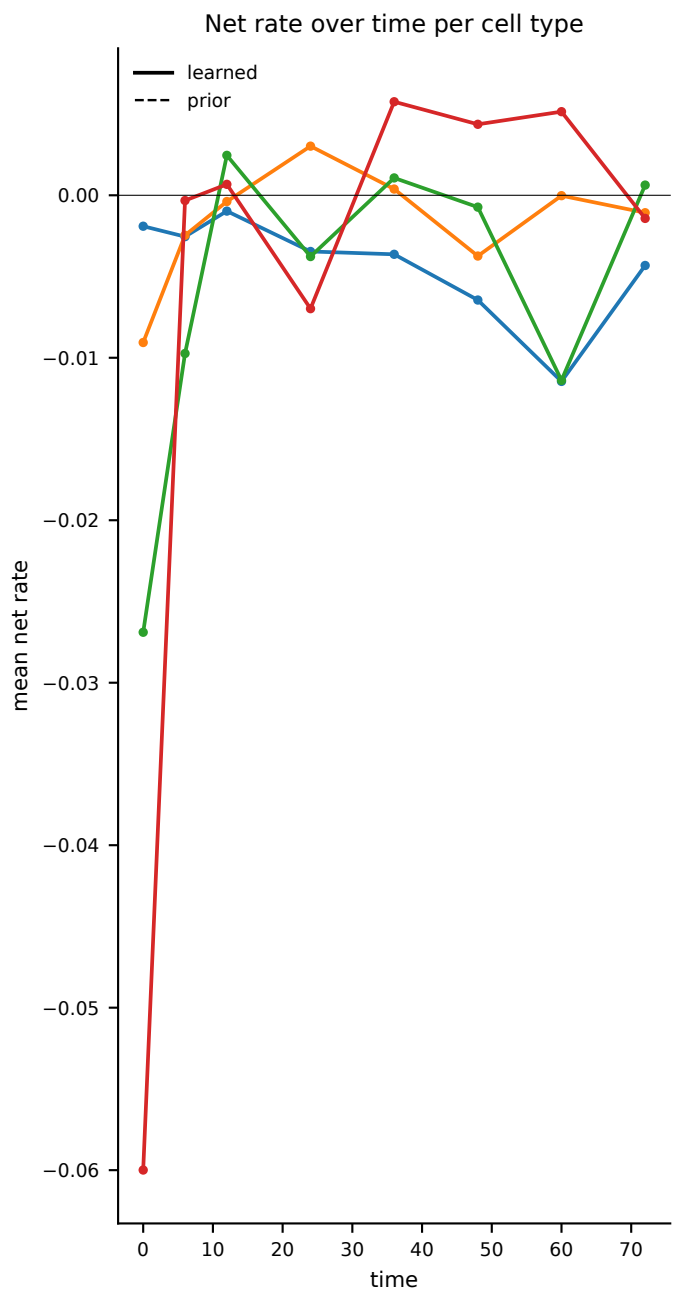
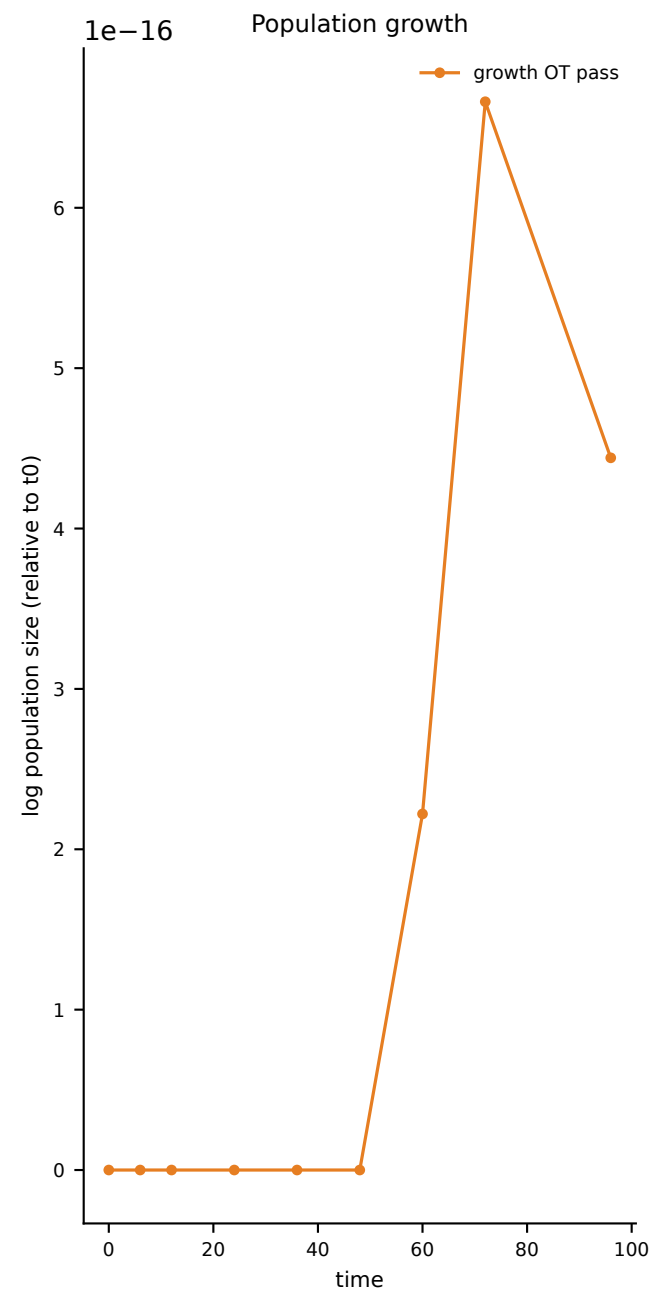


Mean net rate per cell type

Cell type	Anchor	Prior	Learned
Pluripotency	—	—	-0.00265
Post-implantation_Epiblast	—	—	-6.62e-05
Extraembryonic_Endoderm	—	—	-0.00511
Neuroectoderm	—	—	0.00206

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless Data/population_sizes is given, its absolute level follows the prior.

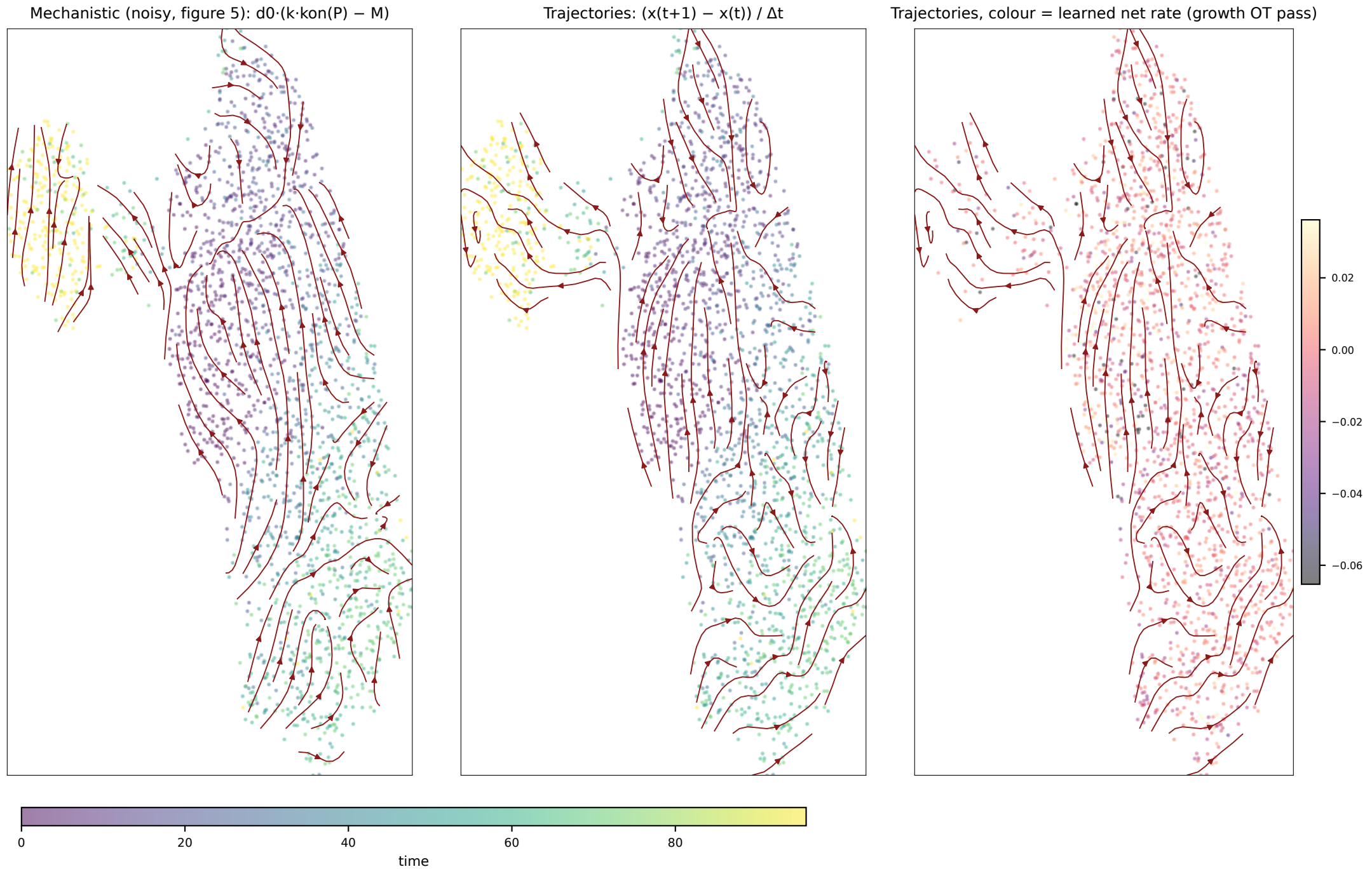


No proliferation MLP
(run infer_network_simul
--compute-proliferation)

Pluripotency Post-implantation_Epiblast Extraembryonic_Endoderm Neuroectoderm

5. Learned dynamics — mRNA velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of log1p NB-sampled mRNA. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.59.



5. Learned dynamics — Proteins velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of protein levels. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.70.

